



FinanceAI Use Case Overview

June 2024

From Use Cases to End-to-End Finance Transformation

Moving from one-off use cases...

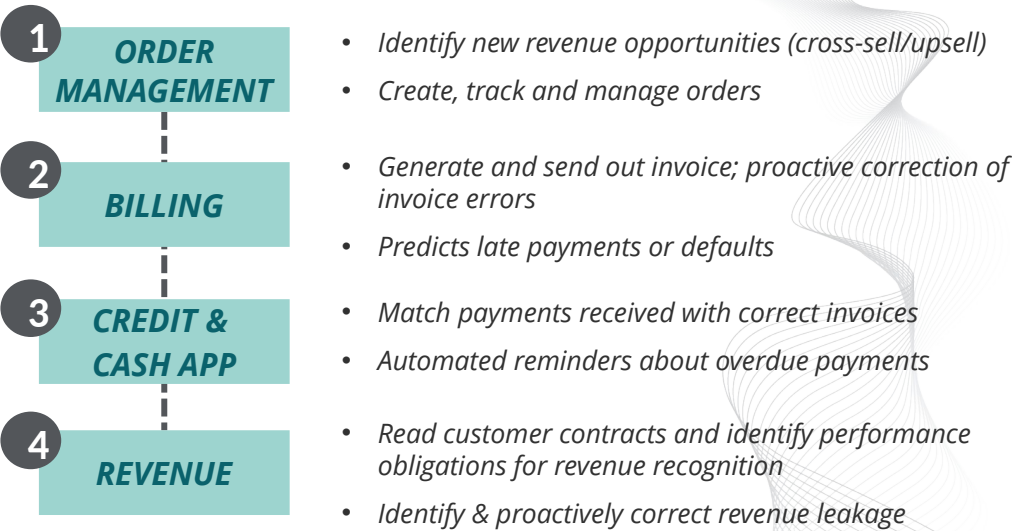
STRATEGIC FINANCE • Macroeconomic analysis • Competitor analysis • Capital allocation optimization and scenario analysis • Corporate development • M&A due diligence	FINANCIAL PLANNING & ANALYSIS • Variance analysis and trend identification • Predictive planning and forecasting • Scenario modeling • Benchmarking analysis (e.g. SG&A, overheads, technology spend) • Financial Data engineering	BUSINESS UNIT FINANCE • Revenue and profitability analysis • Deal desk analytics & insights • Contribution profit modeling • Product costing • Demand forecasting (e.g. product & services bundling) • Monitor and adjust prices • Returns, refunds, chargebacks trending
TRANSACTIONAL FINANCE • Manage PO/invoice matching and payments • Manage billing, collections and revenue leakage • Shared services knowledge support • Predictive risk modeling • Virtual collections assistants • Cognitive spend analysis	CONTROLLERSHIP • Automate accruals calculations from open POs • Prepare internal/external financial reporting including flux analysis, commentary and insight generation • Complete data entry and reconciliations • Source-to-target COA mapping • Intelligent contract review	INTERNAL AUDIT • Identify potential risks and detect fraud • Generate internal audit reports • Monitor compliance with regulations and internal policies
TREASURY • Automate cash forecasting, cash management and provide better visibility into cash flows • Report FX and currency positions, perform counterparty risk and analysis • Advise on optimal mix of investments for liquidity need	TAX • Automate tax preparation and reporting • Interpret new tax laws and provisions • Monitor compliance with tax regulations • Predictive taxability analysis (e.g. Property tax, Sales & Use)	INVESTOR RELATIONS • Develop investor reports and communications • Interpret analyst questions and prepare responses • Valuation and competitive intelligence • Interpret and advise on disclosure rules and regulation

Challenges With One-Off Use Cases

- × **Use-case scope is narrow** and **targets functions in silos**, thereby limiting **benefit realization across Finance**
- × **Higher cost of ownership** for one-off technical efforts

...to INTEGRATED and HOLISTIC capabilities.

Illustrative AI Quote to Cash Solution



Advantages of end-to-end AI Solutions

- ✓ **Broader scope** across Finance provides opportunity to deliver **higher value transformative solutions**
- ✓ **Provides greater return on investment** for client

As part of our Global Use Case Repository we will begin to align to the end-to-end solution structure.

Finance Use Cases (1 of 2)



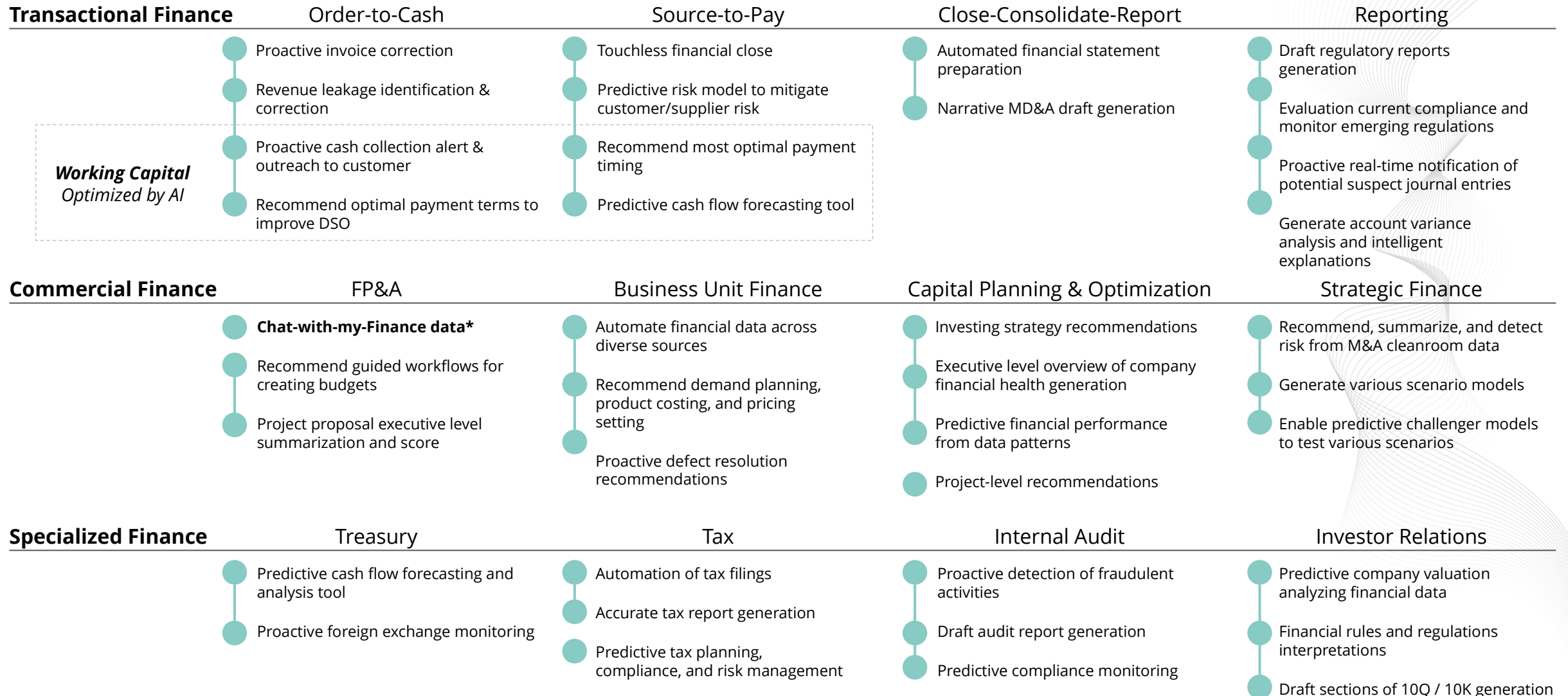
Process Area	Use Case Description	Capabilities	AI Technology	Finance Functions Impacted	
Quote-to-Cash	Invoice and Collections Standardization	• Automated invoice error detection & correction	• ML Predictive + GenAI	• AR • Revenue Accounting • GL	• Transactional Finance • Tax
		• Generate customized invoices for nonstandard customer orders	• GenAI		
		• Facilitate payment collection to identify collections gaps and automate outreach and processing	• GenAI-enabled “virtual assistant”		
Procure-to-Pay	Expense and Profitability Management	• Using data from contract management system, forecast monthly expense for all third-party contracts	• Predictive Analysis	• Expense Management • Revenue Accounting • FRM	• Control & Compliance • Debt & Capital Markets
		• Capture all variations and exceptions in P2P and O2C, enabling a touchless financial close	• AI-enabled process mining		
		• Examine market trends, supplier behavior, and other relevant factors to determine optimal prices, make pricing adjustments, and maximize profitability	• Predictive Analysis + GenAI		
Record-to-Report	Reporting and Compliance	• Generate draft regulatory filings	• GenAI	• Internal Audit	• Tax
		• Monitor emerging regulations and evaluate current state compliance	• NLP		
	General Accounting and Reconciliation	• Monitor potentially suspect or fraudulent journal entries in real time • Generate account variance analysis and intelligent explanations	• LLM • LLM	• Internal Audit • General Accounting	• Controls & Compliance
Working Capital	Cash Forecasting	• Enhance cash flow analysis by minimizing liquidity risks, improving forecasting accuracy, maximizing cash turnaround time and minimizing DSO	• Predictive Analysis	• Treasury Operations • FRM	• Debt & Capital Markets • Investments

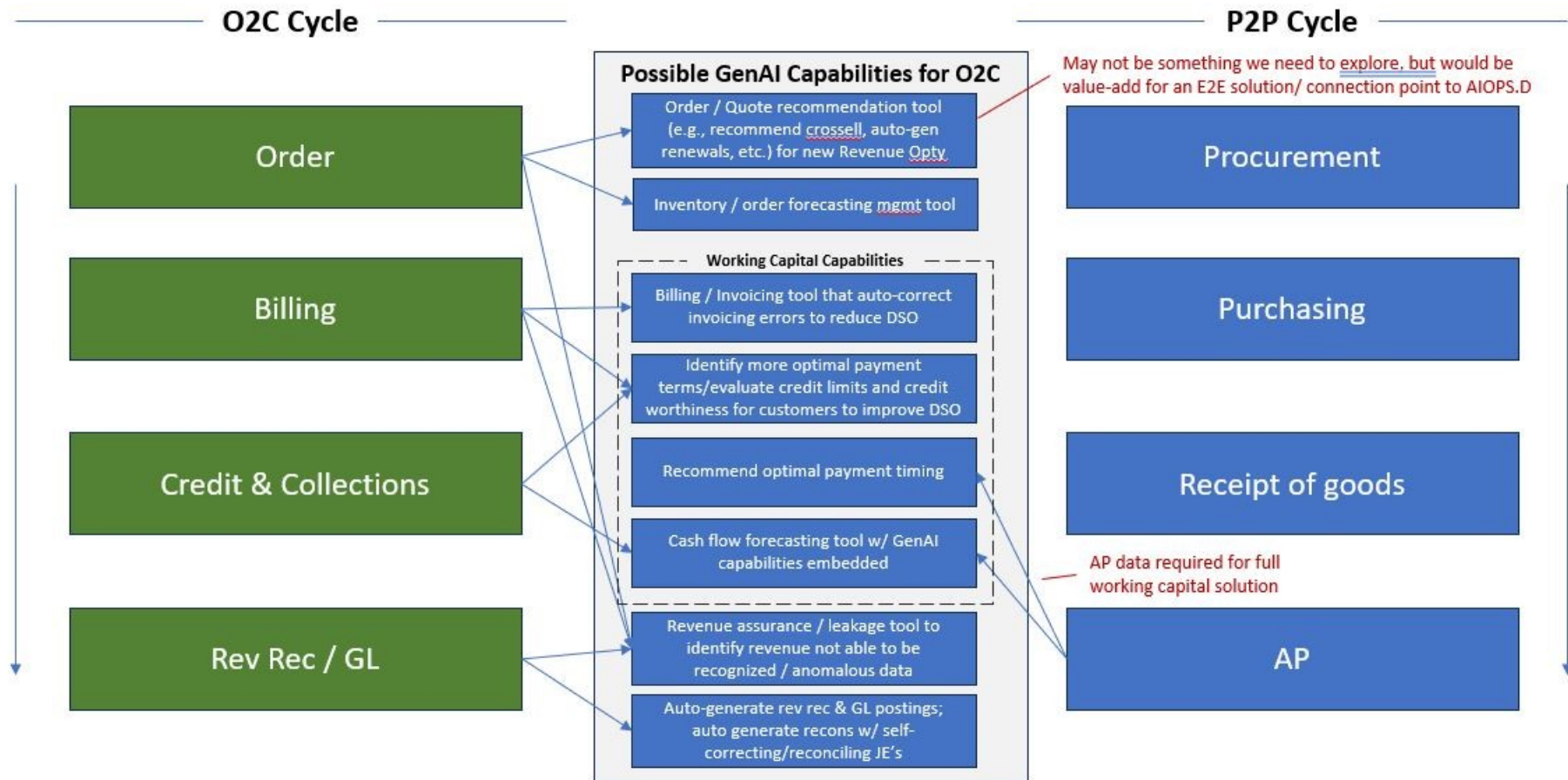
Finance Use Cases (2 of 2)



Process Area	Use Case Description	Capabilities	AI Technology	Finance Functions Impacted	
Capital Project Management	Capital Investment Planning	Generate investing strategies from consolidated market conditions, internal financial performance, and firm objectives	GenAI + LLM	- Strategic Finance - Debt & Capital Markets - Investments	- Forecasting - Insights & Decision Support - BU Finance
		Generate executive level overview of company financial health	NLP		
		Identify past data patterns to predict and improve future financial performance	ML Predictive + GenAI		
Planning, Budgeting, Forecasting	Budgeting and Forecasting	Develop guided workflows for creating budgets that can consider financial metrics and user input	Predictive Analysis + GenAI	- Long-range Strategic Planning - Budgeting	- Forecasting - Insights & Decision Support
			Predictive Analysis + GenAI		
		Summarize and score project proposals across the organization to facilitate executive review			
Treasury	Cash and Foreign Currency Management	Enhance cash flow analysis, minimizing liquidity risks, improving forecasting accuracy, and maximizing returns on cash holdings	Predictive Analysis + GenAI	- Treasury Operations - FRM	- Debt & Capital Markets - Investments
		Optimize foreign exchange operations by evaluating foreign exchange data by identifying patterns and trends to reduce financial risks	Predictive Analysis + GenAI		
Tax Management	Tax Filing and Legal Compliance	Automate tax filing processes, analyze financial data, and generate accurate tax reports	NLP	Tax Accounting & Provision	Direct & Indirect Tax Compliance
		Evaluate tax laws and regulations and provide recommendations	NLP		

From Discrete Use Cases to **End-to-End Finance Transformation with AI**





		CFO Strategy								
Domain		Strategic Finance	Financial planning and analysis (FP&A)	Business unit (BU) finance	Transactional finance	Controllershship	Internal audit	Treasury	Tax	Investor relations
		Finance visioning	Long-range planning	R&D finance	Source-to-pay	Close	Risk assurance	Treasury operations	Tax accounting and provision	Capital markets financing and valuation
Capability		Capital optimization	Budgeting and forecasting	Commercial sales finance	Order-to-cash	Consolidation	Risk advisory	Financial risk management	Direct tax compliance	Regulatory and governance
		Corporate development	Performance management reporting	Manufacturing and supply chain finance	Capital asset management	External and internal reporting	Risk sensing and foresight	Debt and capital markets	Indirect tax compliance	IR planning and strategy
		Finance Transformation	Insights and decision support	Pricing	Project accounting	Financial analysis		Investments	Tax technology advisory	
		Enterprise transformation	Business case evaluation		Subledger close				Tax advisory	
					Operational reporting				Tax controversy and audit support	
Enablers		Process and technology								
		Data and analytics								
		Talent and delivery model								
		Governance, risk, and controls								

Generative AI for Process Automation

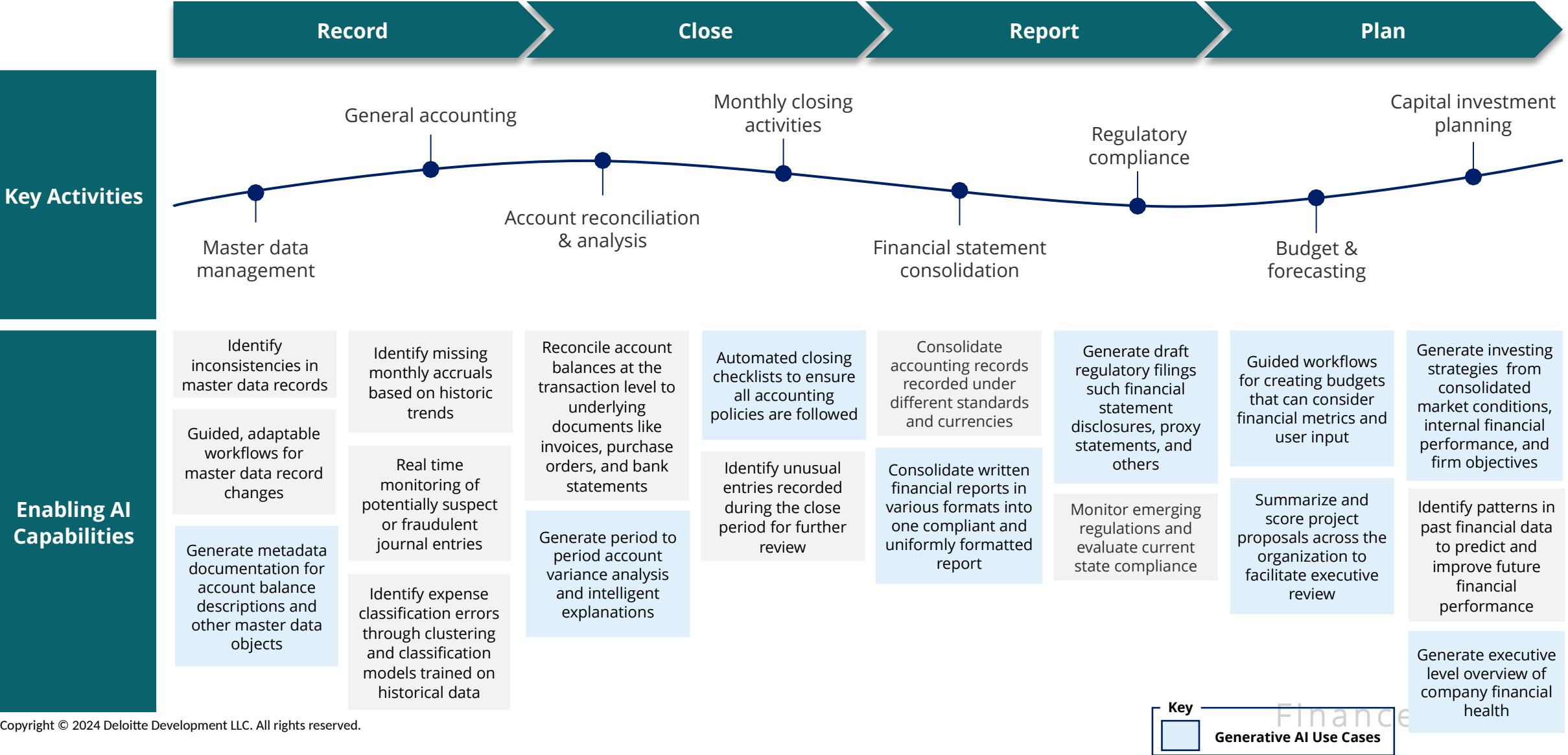
Use Case	 1.1: Minimize / Eliminate Month End Close Processes	 1.2: Contract Lifecycle Management	 1.3: Sales Order Processing	 1.4: Automated & Optimized Stock Replenishment	 1.5: Analytics and Dashboards
Current State	Use the month end process to identify and correct operational issues that occurred during the month. Significant time and resource commitment	Service contracts are written up manually by leveraging templates. Highly manual and error prone , with no ability to automate monitoring compliance	Sales orders are still received manually / calls / emails from customers and entered into SAP by customer service representatives	Stock replenishment is based on daily MRP runs and manual decisions by the planners for each location	Pre-built dashboards for key KPIs , inflexibility in using disparate sources of information to generate real-time insights
Future State	Use GenAI to identify and address transaction processing errors on a real-time basis and minimize / eliminate month end processing	Automate contract authoring, negotiations and sign-off processes using GenAI. Drive to continuous compliance monitoring	Combine GenAI with optical character recognition to fully automate the creation of sales orders in SAP; focus customer service on managing exceptions	GenAI will drive auto stock replenishment across the entire network based on real-time consumer data, demand variability and supply constraints	Dashboards / information is generated real-time based on user questions, independent of the source of the various underlying data

Benefits:

- *Reimagine business processes to not only “execute activities more efficiently” but to rethink “what activities are required”*
- *Reduce End to End Cycle time of business processes, unlocking significant efficiencies*
- *Unlock net new capabilities that were once deemed unachievable (e.g. real-time business performance insights & associated actions)*

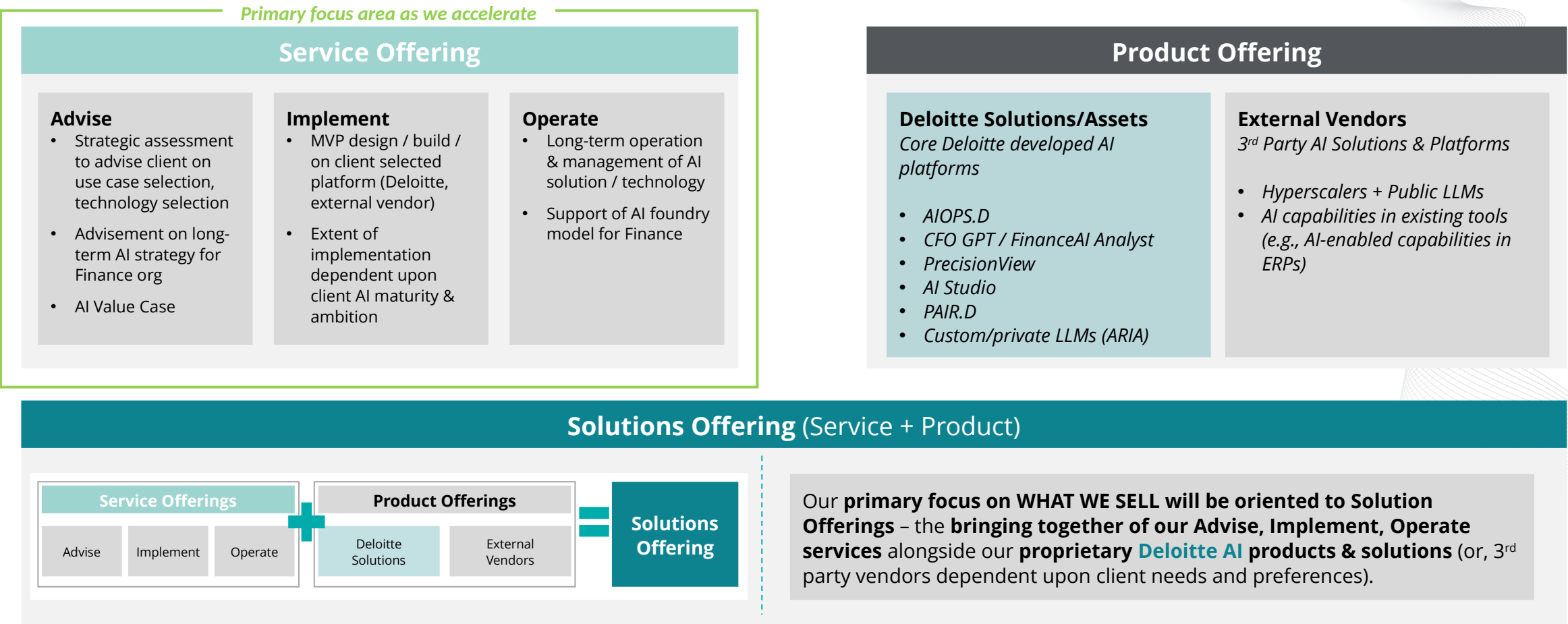
Corporate Finance – AI Use Case Overview & Opportunities for Generative AI

Standardized and repetitive tasks across corporate finance processes are ripe for AI disruption



What We Sell Strategy

We are **focused on selling Solutions that combine our core advise, implement, and operate services alongside AI products** (Deloitte products and/or external products).



Corporate Finance – Generative AI Improvement Opportunity Examples

Transform the way the enterprise engages in capital budgeting and financial reporting by offering generative AI-powered solutions to accelerate planning and development processes

Opportunity	Current State Leading Practices	Generative AI Solution	Marginal Benefits with Gen.
a) Budget & Investment Planning	- Manually review compiled budgets & forecasts - Compile feedback across different verticals and summarize - Modify plan based on feedback	- Generate budget documentation and revision materials based on previous successful budgets - Finetune budget planning documentation with specific requests via text prompts automating the process with human oversight	✓ Prompt based document revisions for specific budgeting requirements ✓ Decrease time spent drafting overhead documentation for leadership review
b) Month End Reporting	- Account level activity is manually reviewed to understand variance drivers and generate relevant commentary - Synchronous, executive level meetings are required to discuss variances	- Generate concise, understandable explanation of cost and revenue drivers of account level variance - A interactive UI could allow natural language inquiries about the variances and reports	✓ Less time spent on aggregating data and diving into account variances ✓ Time efficiencies from fewer synchronous month end close meetings
c) Generate Regulatory Filings	- Semi-automated processes to generate the input figures - Manual process to aggregate financial data and elaborate to conform to regulatory filing needs	- Generate first drafts derived from previous filings, internal financial information, and filing requirements guided by workflows - Highlight potential edits influenced by filings of other companies	✓ Less time wordsmithing initial drafts ✓ Less time spent researching historical filings ✓ Guided workflows act as a completeness check

c) Generate Regulatory Filings *Generate draft filings of various regulatory documentation*

Generative AI can support the process of creating regulatory filings by saving time in the initial drafting stages by learning regulations, workflows, and patterns in previous documents.



Guided workflows provided a structured format to specify the type of filing to generate, the type of information to include, and the desired objective of the filing (i.e. a **proxy statement** that highlights the **equity incentive plan** and the result of the **board of director's** vote)



LLM can identify patterns in historical internal and external filings to generate first drafts, saving time otherwise spent wordsmithing to transition ideas to formatted legal documents



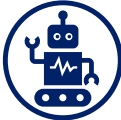
Summary of the filing and considerations taken generated for supplemental review by executive level.

Deep-Dive into Select Gen AI Applications in Finance

	FP&A		Transactional Finance		Controllershship	
	Monthly Reporting	Company Report	Account / Transactional Inquiries	Global Business Services	Report Reconciliation	Regulatory Adherence
Current State Leading Practices	- Perform account level variance analysis - Manually review compiled budgets & forecasts - Compile narrated report for distribution	- Closely monitor regulatory requirements - Manually draft and publish the company reports - Prepare scripts for earnings calls and analyst briefs	- Customer Rep assigned to a small number of customers - Rep efficiency dependent on familiarity with customer's account - No aggregated knowledgebase for resolution	- SOP's and FAQ's are manually documented by experienced resources - Valuable resources deployed performing low value tasks	- Financial documents are manually tied out to underlying support - Financial reports are tied out internally to ensure mathematical accuracy and balancing	- Gap analysis is manually performed as a completeness check against missing disclosures - New regulations must be sorted through and applied or developed into accounting policy
Generative AI Solution	Automatically generated draft periodic management reports – both numeric, format and narrative	- Monitor regulatory changes actively - Automatically generated a draft version of the company report	- Past incident history mined and analyzed to assist in future transactional research - Intelligent self-service chatbot to resolve customer concerns	- New SOP's generated on the fly to capture small changes in processes - Q&A chatbot to guide new employees through SOP's	- Reports are tied out automatically - Variances are identified and potential causes are explained (with drivers / root causes)	- Automated review and summarization of potential gaps in disclosures - New regulation is filtered on applicability and impact is summarized
Value Enabled	✓ Faster Close ✓ Shorter "time-to-insights" cycles ✓ Broader scope of variance analysis					✓ More comfort in disclosure completeness ✓ Ability to proactively plan for reg changes

F&P Priority Use Cases

Six priority use cases have been identified to develop and explore with clients – these will form part of a broader FinanceAI platform concept, and provide necessary experience to evolve our offerings

Use Case	 FinanceAI Insights Platform	 Mgmt Reporting – Actuals, Plan, Variance + Commentary	 Shared Services Interaction Layer - Intake / Automation	 Forecasting & Scenario Modeling – Automated / On Demand	 Journal Entry Automation – <i>Journal Entry Automation</i>	 IR Analyst Augmentation – <i>Scripts, Q&A, Articles</i>
Current State	Significant human time and effort is consumed, analyzing and summarizing financial performance data and reports to provide leaders with actionable insights.	Management reporting is labor intensive with humans connecting datasets, addressing data issues, producing reports, analysis and commentary	Employees interact with shared service functions in various, fragmented and rigid ways using multiple communication mediums.	Forecast accuracy and timeliness limit true effectiveness. Scenario modeling is unavailable or lacks enough functionality to meet business needs	Meetings and emails across teams create a bottleneck and slow the pace of close activities. AJE documentation is fragmented, and results are inconsistent	Financial information and commentary is manually consolidated and iterated from multiple sources and reviewed with business leaders to come up with a script and Q&A plan.
Future State	Perform analysis & investigation, and provide 1 st order insights on demand from multiple financial sources	Produce mgmt reporting & analysis (<i>Version 1</i>) on demand in any medium including insights and commentary	Provide a new conversational AI (chatbot) interaction layer where employees can request any services based on free-form request	Provide end users with on demand forecasting and scenario modeling functionality with complete parameters and drivers flexibility	Automated proposed AJEs are drafted and reviewed with applicable stakeholders in real-time, with limited resource follow-ups, and automated summarized documentation	Automate, accelerate and generate 1st version artifacts for earnings and investors (e.g., Earnings transcript, Q&A prep, articles, 10K/10Q)
	Value H Effort M	Value M Effort H	Value H Effort H	Value M Effort M	Value M Effort H	Value L Effort L

Proof of Concept | FinanceAI Insights Platform

Perform analysis & investigation, and provide 1st order insights on demand from multiple financial sources



Industries:

- Life Sciences and Healthcare
- Financial Services and Insurance
- Tech, Media, and Telecom
- Energy, Resources, and Industrials
- Consumer
- **Deloitte** / Professional Services

Model: **Proposed**



Earliest available environment
in Deloitte (6/19)

Situation

- Significant human time and effort is consumed, analyzing and summarizing financial performance data and reports
- Financial insights can be out-of-date, subjective, or narrowly sourced
- Generative AI can reduce *human-ware* and latency in insight delivery

Business Objectives



- Provide financial analysts access to generative AI capabilities in concert with their financial language and data
- Querying the insights platform would provide forward-looking and current insights, analysis, data and commentary based on the latest financial information of the client
- This capability would drive data-driven decision-making across the client's organization, and support ad hoc analysis and questions

Solution Approach



- Generative AI / LLM enabled solution
- Interaction layer to converse with the financial analyst
- Orchestration layer to manage interactions and access to key data sets
- Generative AI / LLM model to perform analysis, summarization, insight generation, report production, and data provisioning
- Train/tune the model based on client's financial language, context, and parameters
- API into financial applications for data sourcing

Potential Benefits

- **On demand access to information for finance professionals**, and possibly their customers to perform self-service for information and insights
- **Enhanced, economic decision making** leveraging up-to-date financial information and insights
- **Reduced reliance on human-ware** in ad hoc financial reporting and analysis processes
- **Reduced latency for available actionable financial information** for client decision-making
- **Lower cost to achieve leveraging Generative AI models** vs. traditional automation approaches
Shared Services Interaction Layer – Intake / Automation

Proof of Concept | Mgmt Reporting – Actuals, Plan, Variance + Commentary production

Produce management reporting & analysis (*Version 1*) on demand in any medium including insights and commentary



Industries:

- Life Sciences and Healthcare
- Financial Services and Insurance
- Tech, Media, and Telecom
- Energy, Resources, and Industrials
- Consumer

Model: TBD on client



Situation

- Finance business partners spend more time evaluating data than generating insights
- Developing goal seek scenarios is a time consuming and labor-intensive process
- Even for relatively simple level 0 tasks, multiple stakeholders need to be involved in gathering accurate information

Business Objectives



- Produce first-draft insights and commentary on demand
- Conduct variance analysis and/or data reconciliation between multiple systems automatically
- Generate automated data visualizations with supporting commentary
- Recommend paths for different business management decisions
- Autonomous self-help and user-retrieved information based off prompts

Solution Approach



- Gather integrated financial/operational datasets from known data sources
- Connect data sources and LLM to an orchestration layer
- Define terminology necessary for LLM to understand the datasets
- Train model on correct terminology and update
- Generate data visualizations and corresponding commentary from the orchestration layer
- Highly dependent on client data maturity, common financial language, and standardization

Potential Benefits

- **Enhanced and complete data analysis** enabled by generative AI models that can analyze large volumes of data in short timeframes
- **Fully-autonomous management reporting** from data collation/reconciliation to commentary and insight creation
- **Enhanced decision support** leveraging virtual finance assistants to inform managers with data-driven decisions
- **Reduced headcount** necessary to create reports and conduct analysis

Proof of Concept | Shared Services Interaction Layer – Intake / Automation

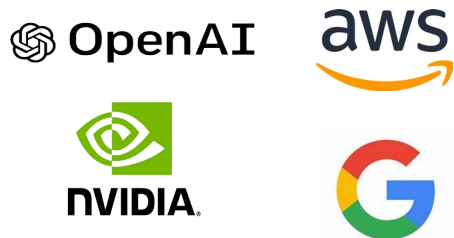
Provide a new conversational AI (chatbot) interaction layer where employees can request any services based on free-form request



Industries:

- Life Sciences and Healthcare
- Financial Services and Insurance
- Government and Public Services
- Tech, Media, and Telecom
- Energy, Resources, and Industrials
- Consumer

Model:



Situation

- Shared services customer experience is not consistent due to multiple channels of engagement and limited self-service capability
- Knowledge articles are stalled, not personalized and not easy to identify
- Shared services customers have very limited visibility into case status, resulting in ~30% increase in call volume

Business Objectives



- Design persona-based customer engagement model to deliver personalized, consistent experience
- Simplify workflows and interaction models leveraging technology, automation and AI
- Exploit data-driven insights, analytics and dashboard to drive performance and improve customer retention

Solution Approach



- Design Persona-based Interface and Workflow based on user journey and the moments that matter
- Define customer intake channel strategy and refresh knowledge article based on customer persona and journey map
- Integrate AI capability and machine translation to customer engagement channels virtual agent (chatbot) that can provide personalized services based on persona and defined knowledge articles to improve self-service and triage/first call resolution and status dashboard
- Track user feedback and volumetrics to inform continuous improvement and data driven customer retention strategy

Potential Benefits

- **Efficiency:** 25-40% productivity gains¹ through optimizing technology stack and digitizing workflows powered by AI
- **Effectiveness:** 4-5x ROI in workflow efficiency through streamlined and effective cross functional collaboration
- **Experience:** 2x increase in customer satisfaction and 25-50% call deflection in call volume through intelligent self-service and coordinated service management

Proof of Concept | Forecasting & Scenario Modeling – Automated / On Demand

Provide end users with on demand forecasting and scenario modeling functionality with complete parameters and drivers flexibility



Industries*:

- Life Sciences and Healthcare
- Financial Services and Insurance
- Tech, Media, and Telecom
- Energy, Resources, and Industrials
- Consumer

Model:

 OpenAI

 AWS

 NVIDIA

 G

Situation

- Finance teams spend significant amounts of time during critical periods to produce and analyze scenarios requested by finance executives
- Merging internal and external datasets (especially operational vs. financial) to accurately generate scenarios is difficult and time consuming
- Manually developing a detailed forecast can take too long to be useful

Business Objectives



- Quickly generate financial forecasts based on real time business scenarios
- Increase the speed of delivery for FP&A modeling
- Utilize the current forecasting toolset to create scenarios that match plan/budget
- Generation of real-time suggested actions based on the sync of daily financial and operational data

Solution Approach



- Identify: Determine the variables to forecast or model
- Collect & Clean Data: Gather and preprocess historical data
- Build & Train Model: Develop and train a predictive or generative AI model
- Validate & Adjust: Test the model on unseen data and refine as needed
- Implement & Monitor: Deploy the model, monitor performance, and make updates
- Iterate: Continually revise and improve the model with new data and insights

Potential Benefits

- **Improved efficiency** due to fewer FTEs required; faster results
- **Deliver insight** by iteratively interacting with a GenAI model to quickly describe, evaluate and modify scenarios with real data
- **Drive transparency** by Identifying/understanding various assumptions and outcomes of scenarios
- **Use common-language** to automatically generate financial drivers that feed financial models

*Non-Exhaustive

Proof of Concept | Journal Entry Automation

Automating the creation of the journal entry processes leveraging generative AI to engage and gather details used to develop journal entries that would normally occur via emails or close meetings.



Industries*:

- Life Sciences and Healthcare
- Financial Services and Insurance
- Tech, Media, and Telecom
- Energy, Resources, and Industrials
- Consumer

Model:



Situation

- Closing timelines are tight and exasperated with emails and meetings to complete tasks to adjust, consolidate, and explain variances across the business
- Limited calendar openings that place stress on close milestones and compliance
- Inconsistency and siloed systems, metadata and business changes restrict the flow of information needed to be reflected in internal and external reporting

Business Objectives



- Automate the information gathering requirements to prepare AJEs (e.g., accruals for GR/IR, contracts spanning multiple periods)
- Spend more time analyzing variances from prior periods
- Automate data gathering, increase productivity
- Accounting to be able to accommodate the increasing complexity as the business grows
- Increased transparency to auditors
- Better financial insights and decision-making

Solution Approach



- Train a Gen AI model to collect historical patterns, closing activities and stakeholders
- Define business and functional stakeholders to engage for closing activities
- Clean and preprocess input data
- Engage stakeholders to develop and document AJEs and narratives for reporting
- Summarize, document, and send proposed AJEs to applicable Accounting team

Potential Benefits

- **Quicker time to close**
- **Documented support**, through the AI generated conversations with business and functions
- **Reduced workload** for key resources during monthly/quarterly close
- **Foundational data gathering for reporting** and variance inferences
- **Reliable audit trail**, enabling better traceability
- **AJE validation meetings are minimized**, allowing the team more time to analyze and interpret results

*Non-Exhaustive

Proof of Concept | Investor Relations Analyst Augmentation

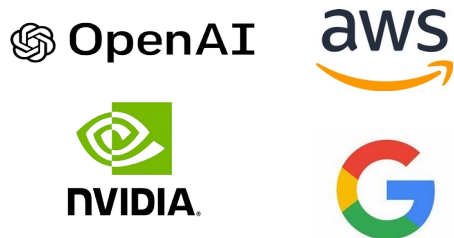
Automate, accelerate and generate 1st version artifacts for earnings and investors (e.g., Earnings transcript, Q&A prep, articles, 10K/10Q)



Industries:

- Life Sciences and Healthcare
- Financial Services and Insurance
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- Consumer

Model:



Situation

- Investor relations (IR) can be complex and time-consuming, with high expectations and difficult stakeholders
- Many core investor relations activities fall within a short time frame, making automation very valuable
- Earnings calls and the associated materials are a critical interaction between the company and its investors

Business Objectives



- Utilize Generative AI to produce the script for the ~5-minute financial update on an earnings call
- Script will include basic narratives such as period over period changes, business updates, high-level financial analysis (from the 10Q)
- The script will also incorporate industry specific transcript credentials, grounded in the analysis of key competitor data

Solution Approach



- The T-5 model facilitates the generation of articles based on key facts and figure provided
- The model further provides top X number of processed and filtered articles for POC based on rough L-Score
- The model is fine tuned to generate articles for further investor communication
- As next steps, manual amendments can be made to the output generated by OpenAI LLM model

Potential Benefits

- **Improve efficiency** by automating routine tasks and freeing up time of investor relations team to focus on strategic tasks such as developing communication strategies
- **Improve consistency** by using GenAI to ensure that communications are consistent in their messaging and formatting, reducing the risk of errors or confusion, and improving the quality of investor communications
- **Ensure compliance** with relevant regulations and standards, and reduce legal and regulatory risks to help improve trust among investors